SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Architecture Design Assignment

Instructions to Students:

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.
- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and	Diagram is incomplete, unclear, or

	major components, interfaces, and interactions accurately represented.		lacks sufficient detail.	technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope; report needs improvement.	Justification is weak or absent; report lacks organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25

AI-Based Fake Identity and Document Screening System

1. System Requirement Analysis (30%)
1.1 Introduction

The AI-Based Fake Identity and Document Screening System is a software application designed to identify fake identities and fraudulent documents using Artificial Intelligence (AI), Machine Learning (ML), and Optical Character Recognition (OCR). It helps organizations verify the authenticity of documents such as identity cards, certificates, and other official records.

1.2 Problem Statement

Manual document verification takes more time and may lead to human errors. Fake documents can be used for identity fraud, unauthorized access, and illegal activities. Therefore, an automated system is needed to detect suspicious documents quickly and accurately.

1.3 Functional Requirements

- Users can register and log in securely.
- Users can upload identity documents for verification.
- The system extracts text using OCR technology.
- AI analyzes document details and detects suspicious patterns.
- The system checks for mismatched information and possible alterations.
- The system generates a verification report.
- Administrators can review flagged documents and make final decisions.

1.4 Non-Functional Requirements

- Security: Protect personal information and uploaded documents.
- Accuracy: Minimize incorrect detection results.
- Performance: Process documents within a reasonable time.
- Scalability: Support multiple users and documents.
- Usability: Provide a simple, user-friendly interface.
- Reliability: Maintain consistent operation and secure records.

2. Selection of Software Architecture (25%)

2.1 Selected Architecture: Layered Architecture

The system uses a Layered Architecture because it separates the user interface, application logic, AI processing, and data storage into independent layers.

2.2 Layers of the Architecture

1. Presentation Layer
Web interface, login, document upload, results display
2. Application Layer
Authentication, validation, workflow management

3. AI Processing Layer
OCR, image analysis, anomaly detection, risk scoring
4. Data Layer
Encrypted document storage, database, audit logs

2.3 Advantages of Layered Architecture

- Easy to develop and maintain.
- Each layer has a separate responsibility.
- Security controls can be applied at different layers.
- AI models can be updated independently.
- Testing and debugging become easier.
- The system can be expanded as requirements grow.

2.4 Alternative Architectures

- Monolithic Architecture: Simple to build initially, but harder to scale individual functions.
- Microservices Architecture: Supports independent scaling but increases deployment complexity.
- Layered Architecture: Offers a good balance of simplicity, maintainability, and modularity for this project.

3. Architecture Diagram and Component Design (20%)

3.1 System Architecture Diagram

User / Administrator
Web Application
Login
• Upload
• View results
Document Verification Engine
File validation
• OCR
• AI analysis
• Risk score
AI Model
Fraud pattern detection
Database
Records and audit logs
Verification Report
Low risk
• Suspicious
• Manual review required

3.2 Component Design

1. User Authentication Component
Verifies user credentials and provides role-based access to the system.
2. Document Upload Component
Accepts supported document formats and checks file type, size, and validity.
3. OCR Component

Extracts text such as names, document numbers, dates, and other relevant fields from uploaded documents.

4. AI Detection Component

Analyzes document images, text inconsistencies, possible alterations, and suspicious patterns to estimate fraud risk.

5. Verification Component

Compares extracted information with permitted reference records or trusted verification sources when available.

6. Report Generation Component

Generates a report containing the risk score, detected issues, and recommended next steps.

7. Database Component

Stores authorized user details, verification results, and audit records securely.

4. Component Interaction and Quality Attribute Analysis (15%)

4.1 Component Interaction

The system works through the following sequence:

1. The user logs in to the application.
2. The user uploads an identity document.
3. The application validates the uploaded file.
4. The OCR component extracts relevant information.
5. The AI component analyzes the document for suspicious patterns.
6. The verification component checks available reference information.
7. The system generates a risk score and verification result.
8. Suspicious or uncertain cases are sent to an authorized administrator for review.
9. The result and audit record are stored securely.

4.2 Quality Attribute Analysis

Quality Attribute	Implementation
Security	Encryption, secure authentication, access control
Performance	Efficient OCR and AI processing
Accuracy	Validated models and reliable reference data
Scalability	Modular components and scalable processing
Maintainability	Independent layers and reusable modules
Reliability	Error handling, backups, and audit logs
Privacy	Data minimization and restricted document access

4.3 Error Handling

- Invalid files are rejected with a clear message.
- Poor-quality images are sent back for re-upload.
- Uncertain AI results are marked for manual review.
- External verification failures are reported without assuming the document is fake.
- System errors are logged securely for troubleshooting.

Important: An AI risk score is not final proof of fraud. Human review and appropriate verification are required before rejecting an identity or document.

5. Architectural Justification and Technical Presentation (10%)

5.1 Architectural Justification

Layered Architecture is selected because the project requires separate functions for document upload, text extraction, AI-based analysis, verification, and data storage.
It provides a clear structure for development and makes the system easier to test, maintain, secure, and upgrade. The AI detection model can be improved without redesigning the entire application.

5.2 Suggested Technologies

- Frontend: HTML, CSS, JavaScript
- Backend: Python with Flask or FastAPI
- AI/ML: Scikit-learn or PyTorch
- OCR: Tesseract OCR
- Database: MySQL or PostgreSQL
- Security: HTTPS, encryption, and role-based access control

These are proposed technologies; the final selection depends on project requirements and available resources.

5.3 Expected Outcomes

- Faster document screening.
- Early identification of suspicious documents.
- Reduced manual verification workload.
- Consistent verification reports.
- Improved security and traceability.
- Better support for human verification decisions.

5.4 Conclusion

The AI-Based Fake Identity and Document Screening System combines AI, OCR, and secure software architecture to support automated document verification. The layered architecture separates system functions and improves maintainability, security, and scalability. By combining automated analysis with human review, the system can help organizations identify suspicious documents more efficiently while reducing the risk of incorrect decisions.